CYBER SECURITY × GENERATIVE AI

# AI is sharpening cyber attacks. We sharpened the defence.

We built an **AI security analyst** that investigates intrusion alerts and **writes its own detection rules** to catch attacks in network traffic, then honestly measured how well it does the job.

*Automated Generation of Intrusion Detection Rules Using Generative AI in a Rule-Based IDS Deployment*

**13/14**
attack types identified by the AI analyst
CIC-IDS2017 dataset

**99.4%**
of its rules' alerts were real attacks
stock rules: 6.1%

**8/9**
attack types found on a dataset it had not seen before
UNSW-NB15 dataset

**0**
false alarms from its rules on unseen normal traffic
held-out test day

## Abstract

Intrusion detection only catches what its rules describe. In our open-source pipeline an AI model plays the **security analyst**: it investigates alerts, then writes and tests rules. We scored it on answer keys it never saw.

## Background

AI is expected to make attacks more frequent and effective (NCSC, 2024), while defenders drown in alerts.

**Our question:** can the same AI do the analyst's job, and what does it need to know?

## Objectives

- **Build** an AI analyst
- **Measure** every rule it writes
- **Test** what context it needs
- **Compare** it with human experts' rules

## Methodology HOW IT WORKS

**1 Traffic** — Recorded attacks from two public datasets →
**2 Suricata IDS** — Open-source engine raises alerts →
**3 AI analyst** — Investigates each alert with 13 tools →
**Threat → new detection rule** — Tested by replay, then deployed
**Harmless → silence the alert** — Only if a **human approves**

### How it became agentic

**Think** plans its next step → **Call a tool** picks 1 of 13 → **Engine checks** then runs it → **Read result** and repeats

↻ Repeats until a verdict (max 100 turns). Hunt rules must stay **quiet on normal traffic**.

| | |
|---|---|
| Investigate | `query_eve_json · count_alerts · count_by · read_file · list_directory` |
| Hunt | `stack_values` |
| Research | `search_rules · lookup_vulnerability · suricata_keywords` |
| Validate | `run_suricata · test_rule` |
| Write | `write_file · append_to_file` |

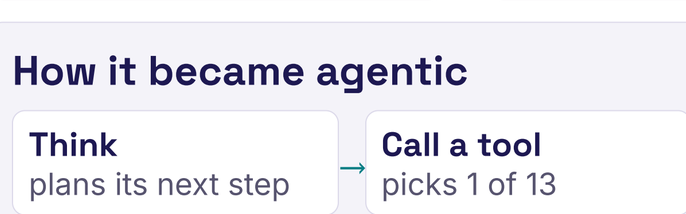
### Tech stack

- **Suricata 8** IDS engine
- **ET Open** + 3 more open rulesets
- **Python** agentic triage engine
- **DeepSeek V4 Pro** & **Claude**, one adapter
- **FastAPI** web UI, streams each step live
- **Local NVD** vulnerability database

## Results PRECISION

### Is the alert a real attack?

CIC-IDS2017, noisiest capture days

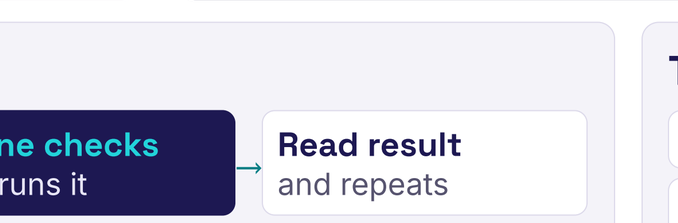



With full context it could tell **if an attack succeeded**: 52 times vs 9.

## Results VS HUMANS

### Gain over a strong human baseline

Unseen test day; baseline caught 29.5%




Hunting like **real detection engineers** worked best, with **zero** false alarms.

## Analyst moments

**Fixed its own mistake**
First rule failed; it found why, fixed it.

**Found what we missed**
Spotted a botnet our report had missed.

**Precise, not noisy**
One rule: 1% → 28%, no false alarms.

## Conclusions

Can AI replace the security analyst? **Our answer is split.**

**Yes, for triage**
Investigates alerts end to end and explains them.

**Partly, for detection**
Real gains on unseen data; risky changes need a human.

## Honest limits

- Recorded traffic, not a live network
- Detects only, does not block
- Made confident mistakes

**Scan to see the AI analyst at work**
`triage.bigyan.work`

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Sharafaldin et al. (2018) CIC-IDS2017 · Moustafa & Slay (2015) UNSW-NB15 · NCSC (2024) The near-term impact of AI on the cyber threat.